100-Question Logic Building Roadmap

Level 1: Basic Input/Output & Arithmetic (Questions 1–15)

1. Focus: Master variables, data types, and basic operators.
2. Swap two numbers using a third variable.
3. Swap two numbers without using a third variable.
4. Convert Celsius to Fahrenheit and vice versa.
5. Find the area and perimeter of a circle, rectangle, and triangle.
6. Calculate simple interest and compound interest.
7. Given a 3-digit number, find the sum of its digits.
8. Reverse a 3-digit number arithmetically (without strings).
9. Convert a total number of days into years, weeks, and days.
10. Find the ASCII value of a character.
11. Check if a given number is even or odd without using the % (modulo) operator.
12. Compute the quotient and remainder of two numbers without using / or %.
13. Calculate the gross salary of an employee given basic salary and allowances (HRA%, DA%).
14. Take a character input and print its next 3 consecutive characters.
15. Find the maximum of three numbers using the ternary operator.
16. Convert a given number of seconds into hours, minutes, and seconds.

Level 2: Conditional Statements & Control Flow (Questions 16–35)

Focus: Mastering if-else, switch-case, and logical operators (&&, ||, !).

17. Check whether a given year is a leap year.
18. Check if a character is a vowel or a consonant.
19. Check whether a character is an alphabet, digit, or special symbol.
20. Check if an alphabet is uppercase or lowercase.
21. Find the largest among three numbers using nested if-else.
22. Input three sides of a triangle and check if the triangle is valid.
23. Classify a triangle as equilateral, isosceles, or scalene based on its sides.
24. Find all roots of a quadratic equation $ax^2 + bx + c = 0$.
25. Simulate a basic calculator using switch-case.
26. Calculate electricity bills based on tiered unit pricing (e.g., first 100 units at $x, next 100 at $y).
27. Determine the quadrant of a coordinate point (x, y).
28. Check if a number is a multiple of both 3 and 7.

29. Accept a student's marks and print their grade (A, B, C, D, F) based on a grading scale.
30. Check if three points (x1, y1), (x2, y2), and (x3, y3) are collinear.
31. Given an age, determine if a person is eligible to vote, work, or retire.
32. Check if a number is a "Harshad Number" (divisible by the sum of its digits).
33. Validate a 4-digit PIN (check length and if it contains only digits).
34. Create a rock-paper-scissors logic for two players using numerical inputs (1, 2, 3).
35. Find the day of the week given a number from 1 to 7 using switch-case.
36. Check if a given character is a printable ASCII character or a control character.

Level 3: Loops & Iteration (Questions 36–60)

Focus: Master for, while, and do-while loops, and loop control statements.

37. Print the multiplication table of a given number.
38. Find the factorial of a number using loops.
39. Generate the Fibonacci sequence up to N terms.
40. Find the Greatest Common Divisor (GCD) / Highest Common Factor (HCF) of two numbers.
41. Find the Least Common Multiple (LCM) of two numbers.
42. Check if a number is a Prime Number.
43. Print all Prime Numbers between a given range [A, B].
44. Check if a number is an Armstrong Number (e.g., 153 = 1^3 + 5^3 + 3^3).
45. Check if a number is a Perfect Number (sum of proper divisors equals the number).
46. Reverse an integer of any length.
47. Check if a number is a Palindrome.
48. Count the total number of digits in an integer.
49. Find the frequency of each digit in a given integer.
50. Convert a Binary number to Decimal.
51. Convert a Decimal number to Binary.
52. Convert a Decimal number to Hexadecimal.
53. Find the sum of a geometric progression series.
54. Print the first N terms of the series: 1, 4, 9, 16, 25, \dots
55. Evaluate the series: 1! + 2! + 3! + \dots + N!.
56. Print all factors of a given number.
57. Implement a "Guess the Number" game loop that terminates only when the user guesses correctly.
58. Check if a number is a Strong Number (sum of factorial of digits equals the number).

59. Find the sum of all even numbers and product of all odd numbers between 1 and N.
60. Extract the first and last digit of any number and find their sum.
61. Find the Prime Factors of a given number.

Level 4: Pattern Printing (Questions 61–75)

Focus: Visualizing nested loops, outer loops (rows), and inner loops (columns).

62. Print a Solid Rectangle / Square of stars (*).
63. Print a Hollow Rectangle / Square of stars.
64. Print a Right-Angled Triangle of stars.
65. Print an Inverted Right-Angled Triangle of stars.
66. Print a Mirrored Right-Angled Triangle of stars.
67. Print a Pyramid / Equilateral Triangle of stars.
68. Print an Inverted Pyramid of stars.
69. Print a Diamond pattern of stars.
70. Print Floyd's Triangle:
71. Print a Right-Angled Triangle of consecutive repeating numbers (Row-wise).
72. Print a Right-Angled Triangle of alternating Os and 1s.
73. Print a Pascal's Triangle up to N rows.
74. Print an Hourglass pattern using stars.
75. Print a Hollow Diamond pattern.
76. Print a cross / "X" pattern using stars inside a square grid.

Level 5: 1D and 2D Arrays (Questions 76-90)

Focus: Data storage, indexing, traversal, and matrix manipulations.

77. Find the largest and smallest element in a 1D array.
78. Find the second largest element in an array without sorting it.
79. Reverse a 1D array in-place (without creating a new array).
80. Count the total number of duplicate elements in an array.
81. Remove duplicates from a sorted array.
82. Left-rotate or right-rotate an array by K positions.
83. Separate even and odd numbers of an array into two separate arrays.
84. Search for an element in an array using Linear Search.
85. Search for an element in a sorted array using Binary Search.
86. Add and subtract two Mx N matrices.

87. Multiply two matrices (handle the dimension compatibility logic).
88. Find the Transpose of a matrix.
89. Calculate the sum of the main diagonal and secondary diagonal of a square matrix.
90. Check if a matrix is a Symmetric Matrix.
91. Print a 2D matrix in a Spiral Traversal pattern.

Level 6: Basic String Manipulation & Functions (Questions 91-100)

Focus: Understanding character arrays, immutable/mutable strings, and modular code.

92. Count the number of vowels, consonants, digits, and spaces in a string.
93. Reverse a string without using built-in reverse functions.
94. Check if a given string is a Palindrome.
95. Check if two strings are Anagrams of each other (e.g., "listen" and "silent").
96. Find the highest occurring character in a string.
97. Toggle the case of each character in a string (uppercase to lowercase and vice versa).
98. Count the total number of words in a sentence string.
99. Remove all spaces from a string.
100. Write a recursive function to calculate the factorial of a number.
101. Write a recursive function to find the N-th Fibonacci number.

Language Implementation Tips

If you are using C++:

Use #include <iostream> and std::cın /std::cout for basic I/O.

For arrays, practice passing them to functions using pointers or references (e.g., void process(int arr[], intsize)).

O Use std::string from the <string> library for string manipulation.

If you are using Java:

Use the Scanner class (import java.util.Scanner;) for taking user inputs.

Remember that Java strings are immutable.

When doing extensive string modifications (like in Questions 92 or 96),

use StringBuilder or String Buffer to save memory and avoid execution timeouts.

O Make sure all your executable code resides inside a class method (e.g., public static void main(String[] args)).